

neural networks

CSCI
270



		instances					
		1	2	3	4	5	6
features	x_1 (bias)	1	1	1	1	1	1
	x_2 (height)	6.4	6.0	5.5	5.0	6.6	7.0
	x_3 (mass)	120	200	120	250	260	150
outcome	y (disease)	1	0	0	1	0	1

$$\hat{x}_1 = x_1$$

$$\hat{x}_2 = \max(0, -120x_1 + 40x_2 - x_3)$$

$$\hat{x}_3 = \max(0, -45x_2 + x_3)$$



how might we automate
feature engineering?



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		1	2	3	4	5	6
	$\hat{x}_1$ (bias)	1	1	1	1	1	1
	$\hat{x}_2$ (underweight)	16	0	0	0	0	10
	$\hat{x}_3$ (overweight)	0	0	0	25	0	0
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the engineered features
all have the same form



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can we also treat these weights as parameters?



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$$\begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \end{bmatrix} = \max \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \hat{\theta}_{11} & \hat{\theta}_{12} & \hat{\theta}_{13} \\ \hat{\theta}_{21} & \hat{\theta}_{22} & \hat{\theta}_{23} \\ \hat{\theta}_{31} & \hat{\theta}_{32} & \hat{\theta}_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right)$$

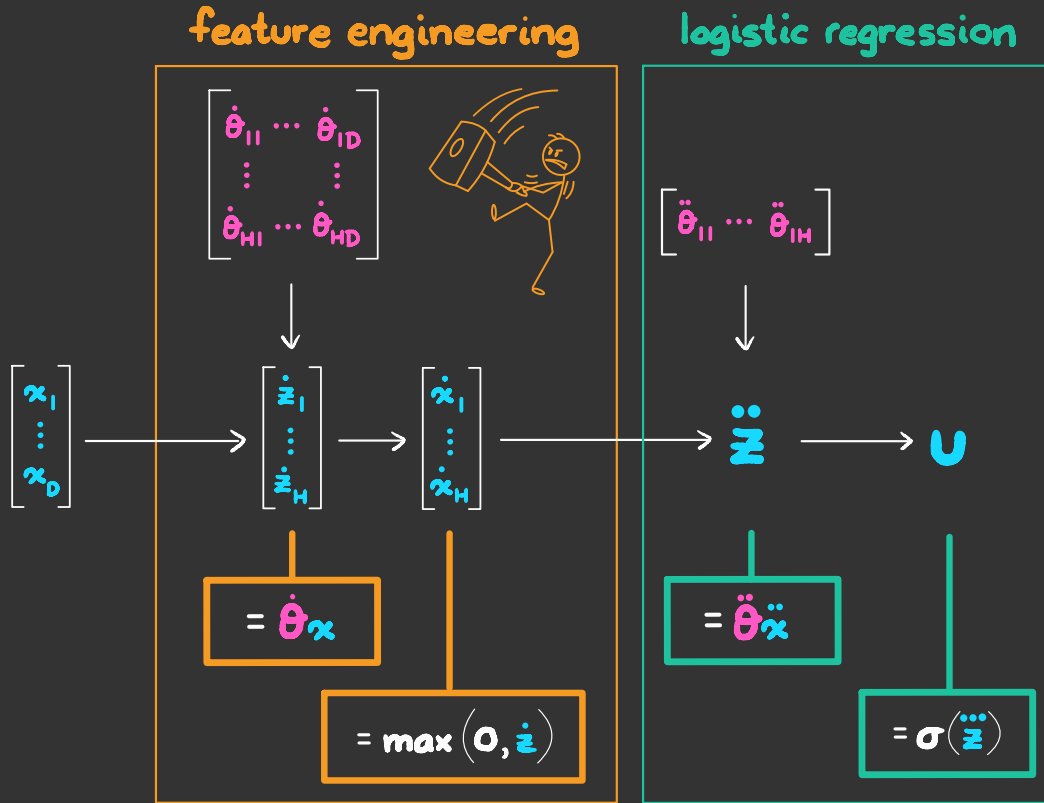


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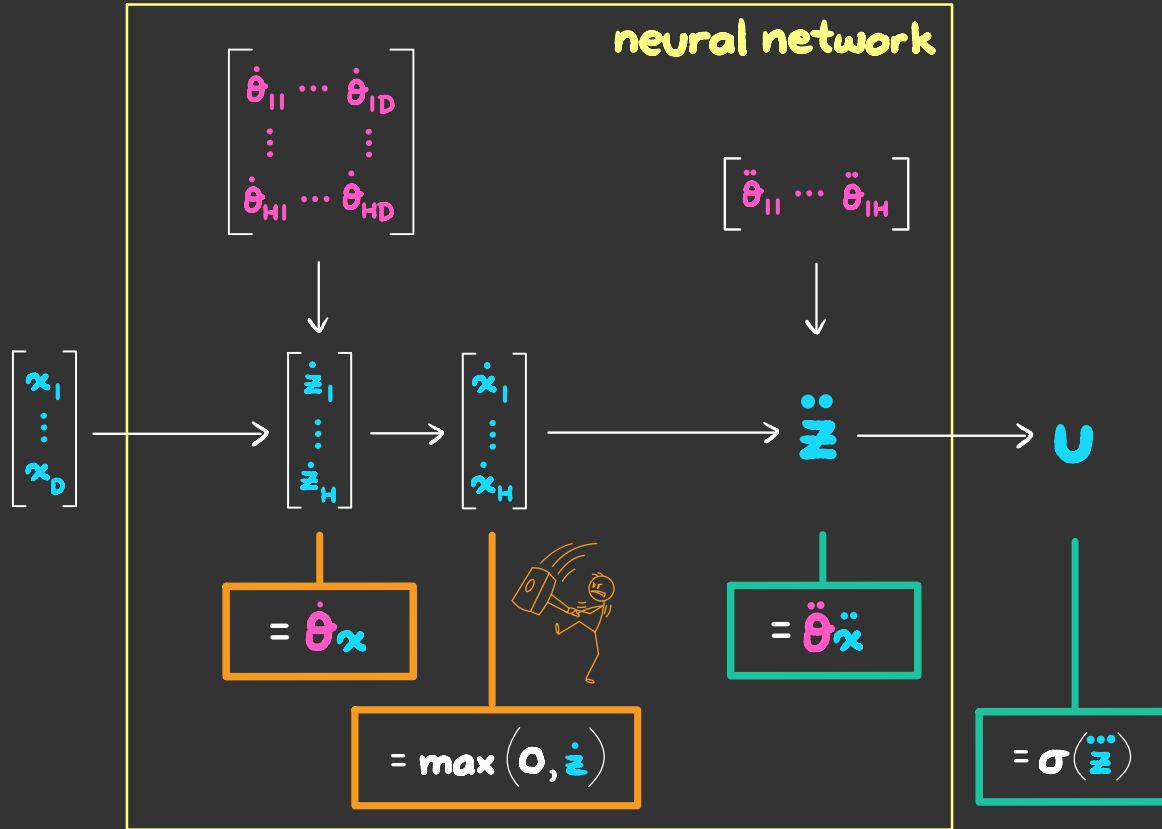


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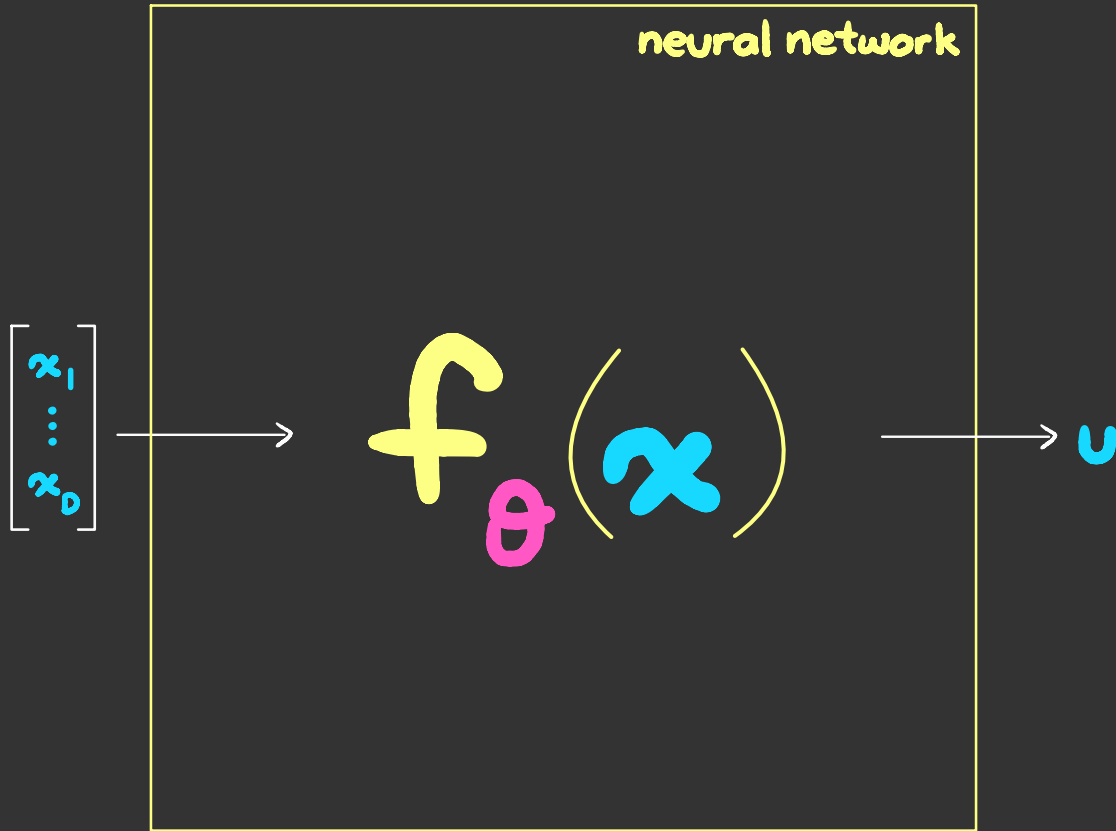
our augmented model



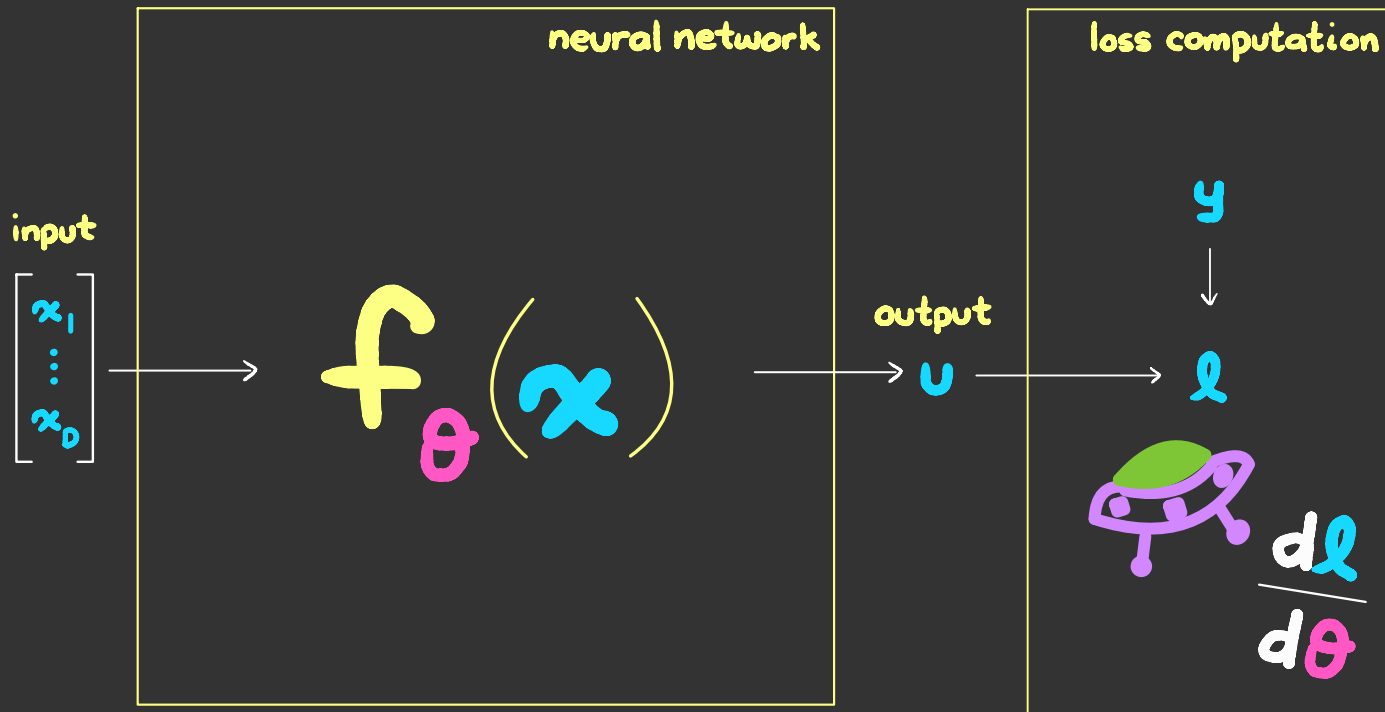
this is called a neural network



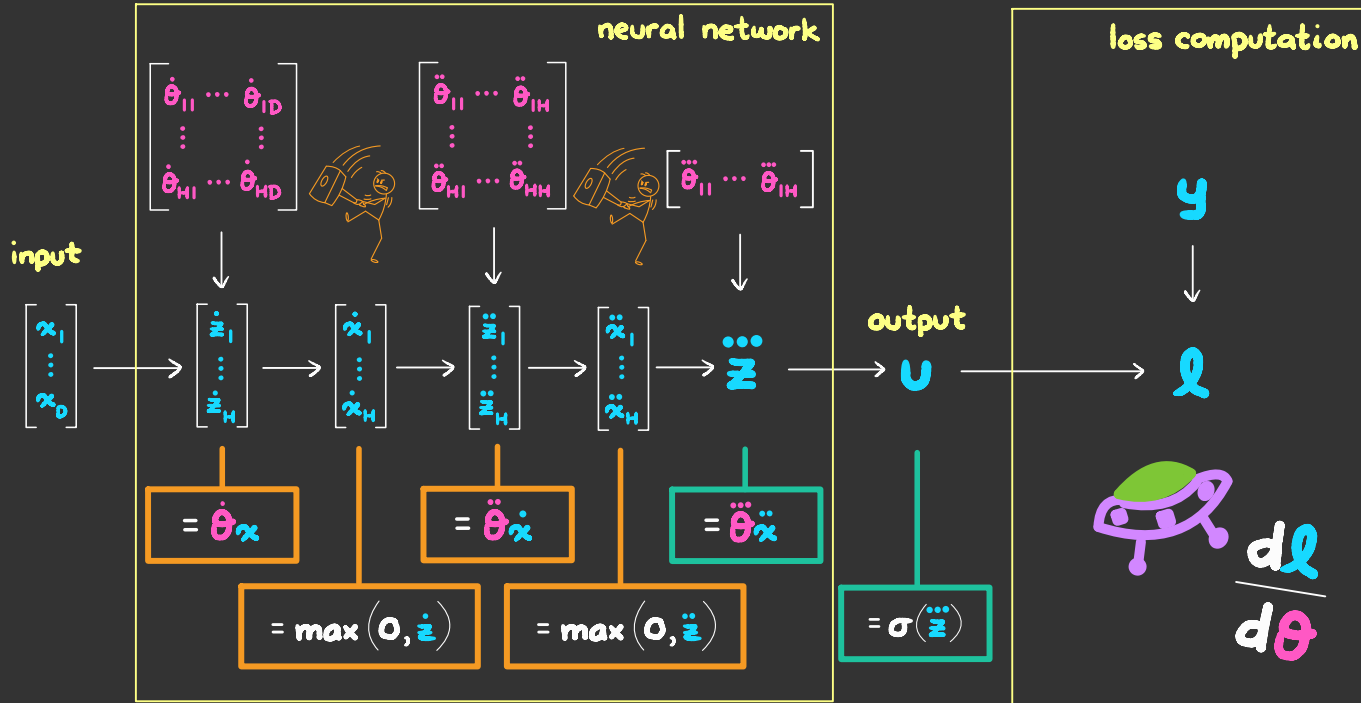
it is still just a parametrized function



and so can be optimized using gradient descent



it can be helpful to perform multiple rounds of **feature engineering**



a multilayer feedforward neural network

